Week 4 — Segmentation Field Guide

Three kinds of segmentation · the four feature families · feature hygiene · choosing k · the three tests · personas and activation · patient segmentation and governance · vocabulary

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Reference handout — keep it next to the notebook and the exercise

Applied to GLUCOREN (6,000 targeted HCPs, 18 months; 20,000 patients); every number quoted is from the executed Week 4 notebook

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How to use this guide. It is the Week 4 analog of the Claims Field Guide (Week 2) and the MMM Field Guide (Week 3): the vocabulary and the checklists a practitioner keeps open while building an HCP or patient segmentation. Parts 1–3 are the method; Parts 4–5 are what you hand to the brand team; Part 6 is patients; Part 7 is vocabulary for interviews.

Part 1 · Three kinds of segmentation

	Decile (volume)	Behavioral	Needs-based (attitudinal)
Built from	One column of Rx volume	Rx panel + claims + CRM/digital activity	Primary research (n = 300–800) + a typing tool
Answers	How big?	How big, how loyal, moving which way, reachable how?	Why? What would change their mind?
Refresh	Monthly	Quarterly / semi-annually; must be stable	Every 2–3 years
Used for	Call frequency, target lists, incentive compensation	Channel mix, cadence, sequencing; the context for NBA	Creative and messaging; persona language
Fails when	Share and trend vary inside a decile (they always do)	Features re-encode volume or activity; unstable across refreshes	Typing tool cannot reproduce the segments on the full universe

The industry answer

Decile for volume, behavior for the play, needs for the message. The three coexist; a segmentation project replaces none of them. Sales operations keeps the decile; marketing gets the behavior; the agency gets the persona.

Part 2 · The four feature families (from claims/Rx and activity data)

Family	Features	Real-brand source	Hygiene rule
Writing behavior	Market volume; brand share; share trend (H2 − H1); NBRx share of brand TRx	Rx panel projections (Week 2)	Log the volume; trend on < 20 brand TRx is noise
Patient mix / access	% Medicare, mean patient age, % CV comorbidity; brand pharmacy-reject rate	Claims (payer, Dx); formulary / reject data	Reject rate is the payer's behavior, not the HCP's — keep it, and route the play to market access
Switching	Switch-in rate (share of NBRx from a competitor); switch-out rate	Longitudinal patient-level Rx (LOT logic)	Switch-out on a small base is a coin flip — drop or shrink

Channel response	Calls received; email open rate; web visits; call response (corr. of brand TRx with last month's calls)	CRM, marketing automation, web analytics	Activity ≠ response; calls encode last year's plan

Feature hygiene — the checklist before scaling

1. **A denominator for every rate.** Below ~20 brand TRx, switch-out rate, NBRx share and call response are noise. Tonight the raw feature set produced a 72-HCP 'segment' with a median of 7 brand TRx and a switch-out rate of 0.17 (universe 99th percentile: 0.12). The k-means objective preferred that solution.
2. **Fix in order of preference:** drop a feature whose only information is the denominator; shrink the rate toward the universe mean (empirical Bayes, weighted by the denominator); winsorize at the 1st/99th percentile.
3. **Log-transform counts** (volume, calls, visits). Pareto-shaped columns dominate Euclidean distance.
4. **Scale, and know that scaling is weighting.** Standardizing gives every feature equal weight; four writing-behavior features and one switching feature means writing behavior gets four votes. Feature-family weights or a PCA per family are the fix when it matters.
5. **Check the correlation matrix.** A pair above |0.5| is one behavior counted twice.
6. **Feature window before outcome window.** Anything you will evaluate on (next quarter's growth) must not be in the features. Tonight: months 1–12 features, months 13–18 outcome.

Part 3 · Choosing k and choosing the method

Three statistics, then the commercial test

- **Inertia (elbow):** always falls with k; the elbow is a judgement. Tonight: around 4–5.
- **Silhouette:** −1…1, how much closer each point is to its own cluster than to the nearest other. Behavioral data: 0.2–0.35 overall is normal. Tonight: 0.28 at k = 3, 0.23 at k = 5.
- **Calinski–Harabasz / Davies–Bouldin:** between/within variance ratio; both prefer fewer, rounder clusters.
- **The commercial test:** can marketing run k different plays? 4–6 is the range. If two segments get the same play, merge them.
- **Two-stage:** k is never known; cluster a value tier first (or use the decile), then behavior within the high-value tier. Tonight: k = 4 single stage ARI 0.92 vs planted; two-stage 0.98.

Methods

Method	Assumes	Gives	Use it when
k-means	Round clusters of similar spread in scaled space; k known	Hard labels; centroids you can explain	The segmentation everyone must understand (default)
Gaussian mixture	Ellipsoidal clusters; k known (BIC)	A membership probability per segment	Boundary HCPs matter (NBA); BIC over-fits long tails — check
Hierarchical (Ward)	Nothing about k; nested structure	A dendrogram; nested segments	Brand wants 'high-value → loyal / contested'
HDBSCAN	Clusters are dense regions; noise exists	Labels + an unassigned set (39% tonight)	Honesty about 'no clear behavior'; hard to sell
Supervised (GBM → deciles)	A label exists (next-period growth)	A ranked score: Champions / Loyalists / Churners	The question is 'who will grow / churn', not 'who behaves alike'

Part 4 · The three tests (in this order)

Test	How	Pass looks like	Tonight
1 · Stability	Bootstrap resamples, other seeds, a shorter feature window; ARI between runs; matched per-HCP agreement	Mean ARI ≥ 0.8 and a minimum not far below; ≥ 90% of HCPs keep their segment	Mean 0.97, min 0.69 (one boundary: core vs movers); 97% agreement; 11-month window 0.997
2 · Separation	Silhouette per cluster; share of negative-silhouette points; Davies–Bouldin	No cluster near zero with many negative points	Overall 0.23; clusters 0.12–0.40; the long tail is the weak one (by construction)
3 · Commercial interpretability	The segment card table: size, brand TRx, headroom, calls per brand TRx, channel response; one play per row	Every row has a different play and an exit trigger	Five plays: protect / grow / fix access / test / cheap digital

Why this order

Naming an unstable cluster is fiction. Separation before interpretability because a leftover cluster should be merged, not named. Report the minimum stability run, not just the mean — it tells you which boundary is fragile, and that boundary is where the second stage or the supervised score goes.

Part 5 · From segments to personas and activation

The segment card (one per segment)

Field	What goes in it	Tonight's example: The Movers (cluster 4 / segment 1.1)
Name	A behavior, never a number	The Movers
Who	Size (% of HCPs), value (% of brand TRx), the two or three features that define them	15% of HCPs · 16% of brand TRx · share 16% rising 8 pts per half-year · half of NBRx switched in · most digital
Needs	What they are looking for (hypothesis until research confirms)	Evidence for switching: head-to-head data, access wins, samples at the moment of the switch
Play	Channel, frequency, message, program	Sequence email → web → call around switching triggers; do not saturate (16.8 calls/year already)
Exit trigger	The observable event that moves an HCP out (the NBA trigger)	NBRx share falls below 30% — they have finished moving
Owner and refresh	Who re-scores and when	Marketing analytics; re-score monthly, re-cluster semi-annually

The activation matrix

Rows = segments, columns = channels, cells = affinity index (the segment's response measure relative to the universe mean; 1.0 = average). Tonight: core and movers respond to every channel (the risk is fatigue → frequency caps); the access-constrained segment is average on email and web but 86% churn risk — the channel is not the problem, coverage is; the entrenched and the long tail are below average on every channel — experiments, not frequency. The matrix, the growth score and the exit triggers are the three inputs to Week 5's Next Best Action layer.

Reconciling with the decile

Sales operations will not give up its deciles, and it should not: deciles drive call capacity and incentive compensation. Present the segmentation as decile × behavior — a two-way table with call frequency in the

decile dimension and the play in the behavior dimension. Two-stage clustering (value tier, then behavior) makes this reconciliation automatic.

Part 6 · Patient segmentation — same machinery, different questions, real governance

- **Features come from the patient journey** (Week 2 pipeline): time from diagnosis to first fill, months on therapy, prior lines, PDC, refill gaps, switches, copay, PSP enrollment, portal logins, comorbidity, ER visits, discontinuation.
- **Questions are different:** who is about to stop, who needs affordability routing, which patients need HCP support. The plays are programs (copay assistance, adherence outreach, nurse support), not messages.
- **Tonight:** k = 4; ARI 0.84 vs planted. Stable long-term (33%, 5% discontinue) · engaged new starters (28%, 55% in PSP) · complex late-line (15%, 3.3 prior lines, 27% discontinue) · cost-sensitive strugglers (23%, copay $69, PDC 0.51, 42% discontinue).

Governance is part of the model

Purpose limitation: a segment built for adherence support is used for adherence support. Never re-purposed for HCP targeting; never joined back to a prescriber list. Label every output with its purpose, refresh date and owner.

Data minimization: only features with modeling utility and low re-identification risk. No names, ZIPs or dates of birth — and none are needed.

Who may act: a patient program may; a sales rep may not know a patient's segment; marketing sees aggregates.

IQVIA's own wording (reading pack, Part D): patient segments 'are labeled with metadata that restricts downstream use to a specific context and time frame.'

Part 7 · Vocabulary

Term	Meaning
Decile	Ten equal-count groups of HCPs ranked by (category or brand) volume; decile 10 = highest
Behavioral segmentation	Clustering HCPs on observed prescribing, switching, patient mix and channel response
Needs-based segmentation	Clustering on surveyed attitudes and needs; scored onto the universe with a typing tool
Typing tool	A classifier that assigns a needs-based segment from available data; the weak link of needs-based work
NBRx	New-to-brand prescriptions; NBRx share of brand TRx measures acquisition vs refills
Switch-in / switch-out	A patient's therapy moving from a competitor to the brand / from the brand to a competitor
Reject rate	Share of a brand's pharmacy claims rejected at adjudication (formulary, prior auth); an access signal
Call response	Correlation of an HCP's brand TRx with the previous month's rep calls — a crude response measure
Winsorize / shrink	Clip a feature at percentiles / pull a noisy rate toward the mean in proportion to its denominator
Silhouette	For each point, (b − a) / max(a, b): a = mean distance to own cluster, b = to the nearest other; −1…1
Adjusted Rand index (ARI)	Agreement between two partitions, corrected for chance: 0 = chance, 1 = identical
Two-stage segmentation	A value tier first (often the decile), then behavior within the high-value tier
Persona	A segment written as a person: name (a behavior), who, needs, play, exit trigger
Activation matrix	Segment × channel table of affinity; the input to Next Best Action
Exit trigger	The observable event that moves an HCP out of a segment; the NBA trigger
Purpose limitation	An output may be used only for the purpose it was built for; labeled with context and time frame

PDC	Proportion of days covered — an adherence measure from pharmacy claims
PSP	Patient support program (hub, nurse, copay assistance) run by the manufacturer